Chapter 3 Practice Quiz Name:

1. Consider the table below that lists positions and times for a cyclist biking East along the Manhan Rail Trail.

t time (s)	9	11	59	61
x position (m)	51	63	467	489

a. Estimate the velocity of the cyclist around $t = 10$ seconds.

b. Estimate the velocity of the cyclist around $t = 60$ seconds.

c. Estimate the acceleration of the cyclist from $t = 10$ until $t = 60$.

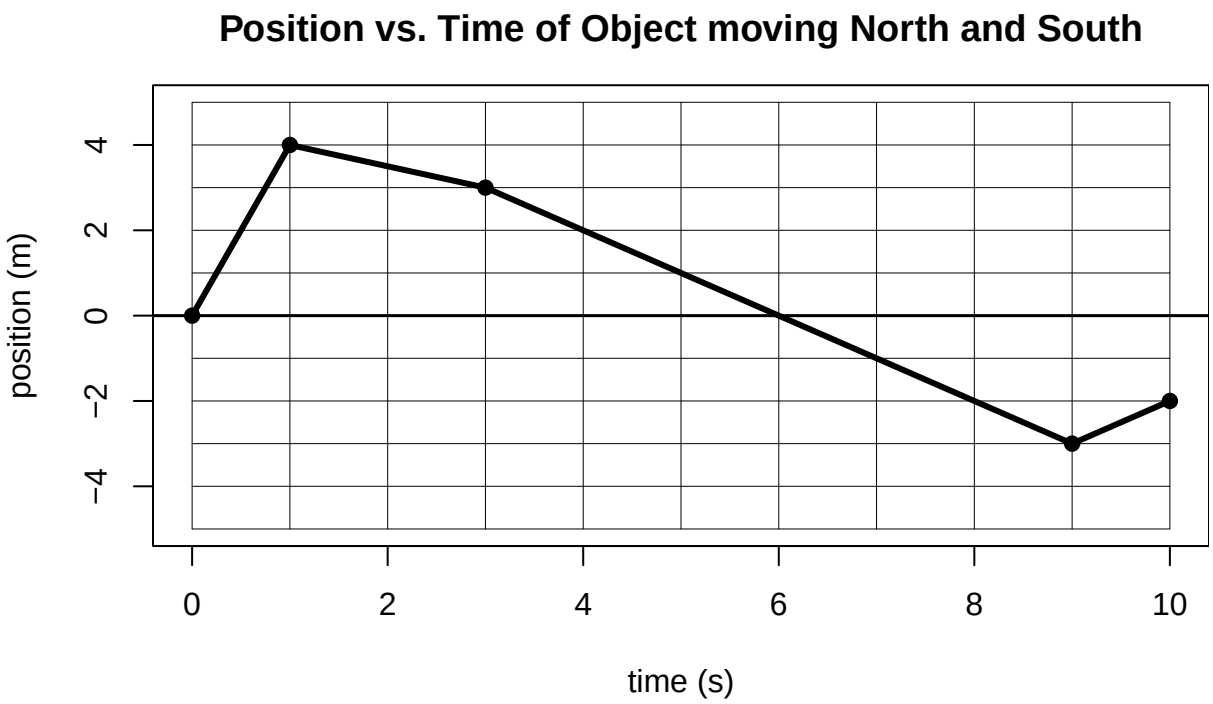
2. A baseball is thrown upward at a speed of 40 meters per second. The acceleration due to gravity is -10 m/s^2.

a. What is the velocity (speed and direction) of the baseball after 2 seconds?

b. What is the velocity (speed and direction) of the baseball after 7 seconds?

c. How long does it take for the baseball to switch directions from going up to going down?

3. Consider the **position** versus time graph shown below. Let positive indicate North and negative indicate South.



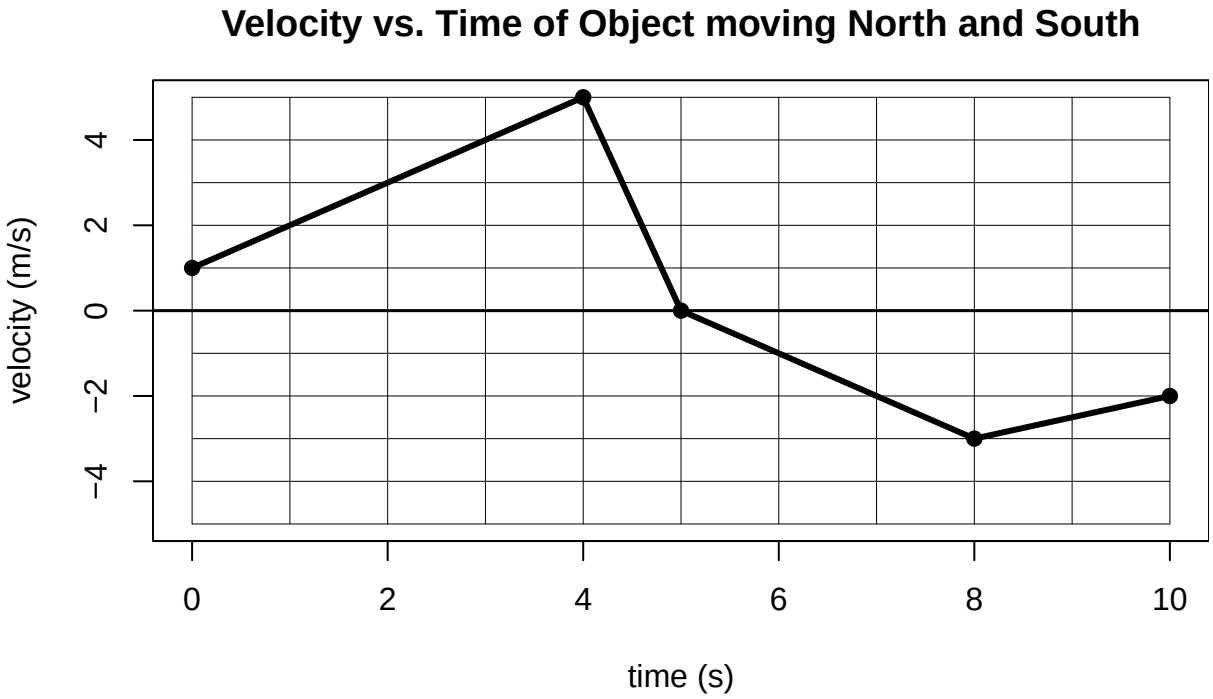
a. When $t = 4$, what is the position of the object?

b. When $t = 4$, what is the velocity of the object? (Remember we need speed AND direction.)

c. During which time interval(s) is the object further North than at $t = 0$?

d. During which time interval(s) is the object moving North?

4. Consider the **velocity** versus time graph shown below. Let positive indicate North and negative indicate South.



a. What is the velocity at $t = 7$ s?

b. What is the acceleration at $t = 7$ s?

c. When does the object change the direction it is heading?

d. During which intervals is the object slowing down (speed getting closer to 0)?

5. A kayaker paddles at a speed of 1.6 m/s relative to water. A river is flowing East at a speed of 1.2 m/s.

a. Relative to ground, what is the velocity of the kayaker when the kayak points East? (Remember we need speed and direction for velocity.) Please draw a sketch to indicate the vector addition and direction of the resultant.

b. Relative to ground, what is the velocity of the kayaker when the kayak points West? Please draw a sketch to indicate the vector addition and direction of the resultant.

c. Relative to ground, what is the velocity of the kayaker when the kayak points North? Please draw a sketch to indicate the vector addition and direction of the resultant.